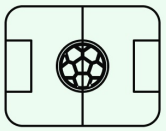
9:41

WhosDown

Find people. Play your game.



1 Create account

2 Sign in

3 Explore as guest

Guests can browse games and groups. Sign in only when you join, post, or message.

H01 — Welcome

1. Auth · createUser(withEmail:password:)
2. Auth · signIn(withEmail:password:)
3. Auth · signInAnonymously → users/{uid}.isGuest = true (AppUser.isGuest)

Endpoints: W1
Other: Auth | **Release: V1**

9:41

‹ Create account

Takes about 30 seconds.

Email

1 alex.morgan@cmu.edu

Password

2 ••••••••••

Display name

3 Alex Morgan

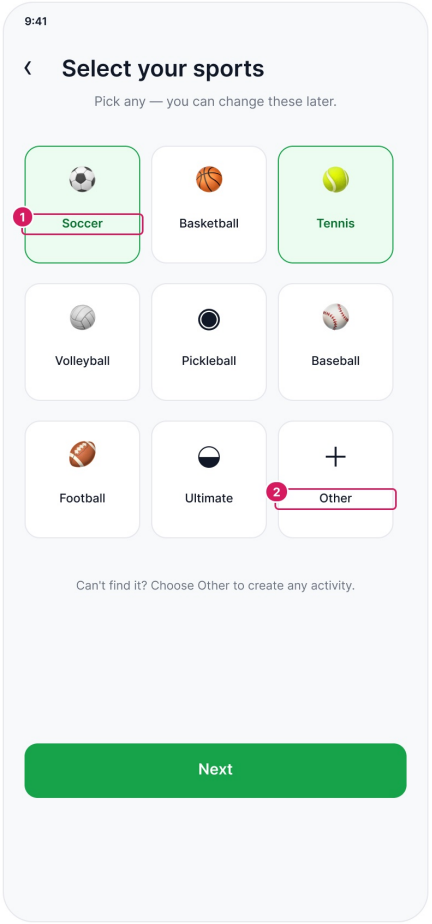
Continue

Already have an account? Sign in

H02 — Create Account

1. Input · Auth email + users/{uid}/private/settings.email (UserSettings.email)
2. Auth · password stays in Firebase Auth only
3. Input · users/{uid}.displayName (AppUser.displayName)

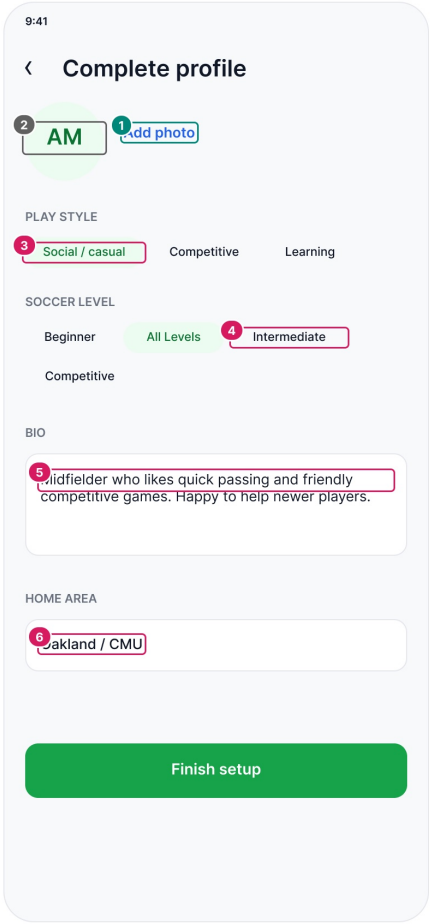
Endpoints: W1
Other: Auth | **Release: V1**



H03 — Select Sports

1. **Input** · users/{uid}.sportLevels keys ([String: SkillLevel], Sport)
2. **Input** · sport "other" → games.customSportName when hosting

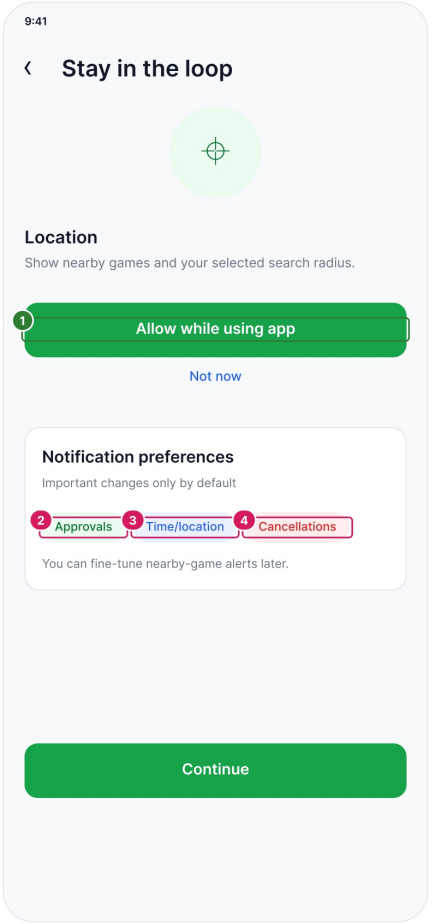
Endpoints: W1
Other: — | **Release: V1**



H04 — Complete Profile

1. **Storage** · Storage users/{uid}/avatar.jpg → AppUser.photoURL
2. **Computed** · AppUser.initials from displayName
3. **Input** · users/{uid}.playStyles ([PlayStyle])
4. **Input** · users/{uid}.sportLevels["soccer"] (SkillLevel)
5. **Input** · users/{uid}.bio
6. **Input** · users/{uid}.homeArea

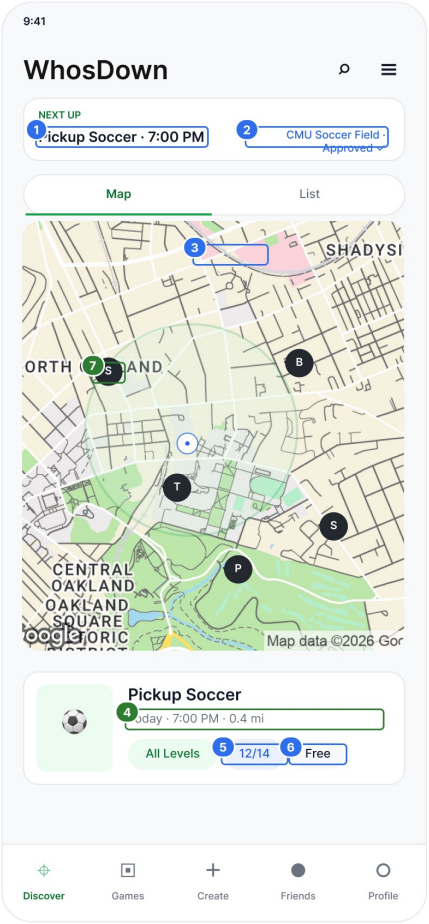
Endpoints: W1 · W2
Other: Storage | **Release: V1**



H05 — Permissions

1. **MapKit** · CLLocationManager when-in-use permission
2. **Input** · private/settings.notificationPrefs.approvals
3. **Input** · notificationPrefs.timeOrLocationChanges
4. **Input** · notificationPrefs.cancellations

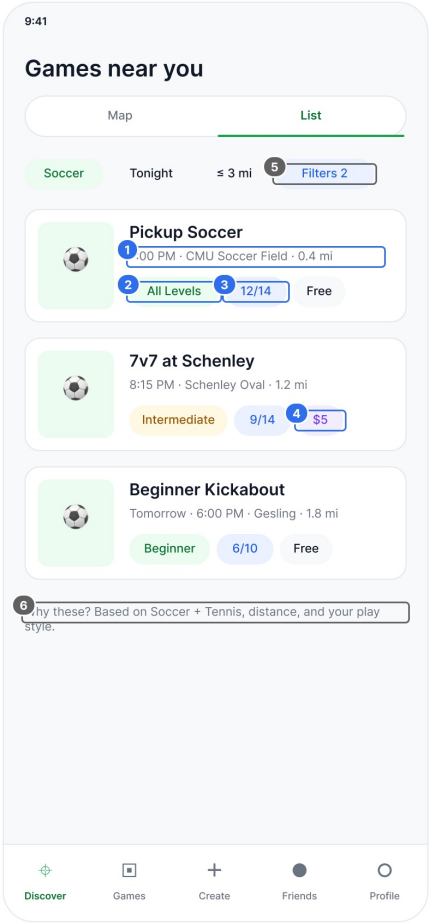
Endpoints: R8 · W1 · W16
Other: MapKit | **Release: V1**



H06 — Discover Map

1. **Firestore** · NEXT UP: collectionGroup rsvps (userId == me, approved) → games.title, startTime
2. **Firestore** · games.venueName + RSVP.status
3. **Firestore** · private/settings.searchRadiusMiles
4. **MapKit** · games.startTime + distance from user location to games.location
5. **Firestore** · games.joinedCount / playerLimit
6. **Firestore** · games.priceCents == 0 (Game.isFree)
7. **MapKit** · Map pins at games.location, icon by games.sport (geohash range query)

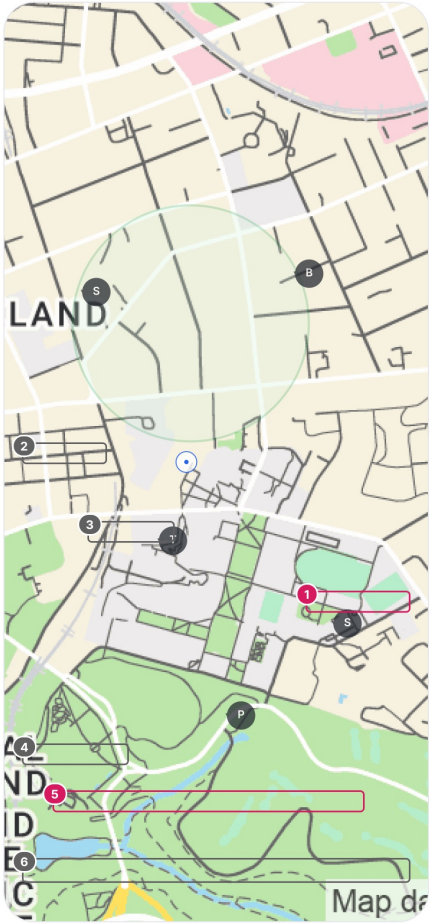
Endpoints: R1 · R5
Other: MapKit | **Release: V1**



H07 — Discover List

1. **Firestore** · games.startTime, venueName + MapKit distance
2. **Firestore** · games.skillLevel
3. **Firestore** · games.joinedCount / playerLimit
4. **Firestore** · games.priceCents
5. **Computed** · count of active filters (view state)
6. **Computed** · AppUser.sportLevels + playStyles + distance

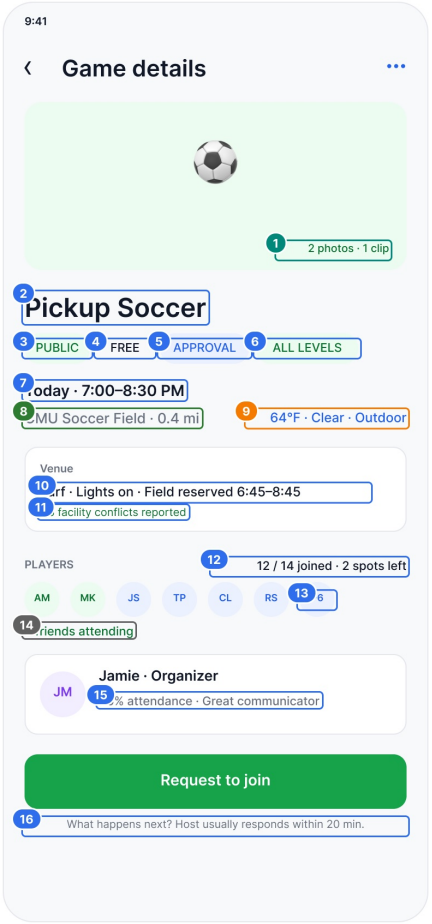
Endpoints: R1
Other: MapKit | **Release: V1**



H08 — Filters Sheet

1. **Input** · private/settings.searchRadiusMiles
2. **Computed** · filter on games.sport
3. **Computed** · filter on games.startTime
4. **Computed** · filter on games.joinType / waitlistEnabled
5. **Input** · private/settings.newGameAlert (NewGameAlertRule)
6. **Computed** · count of filtered query results

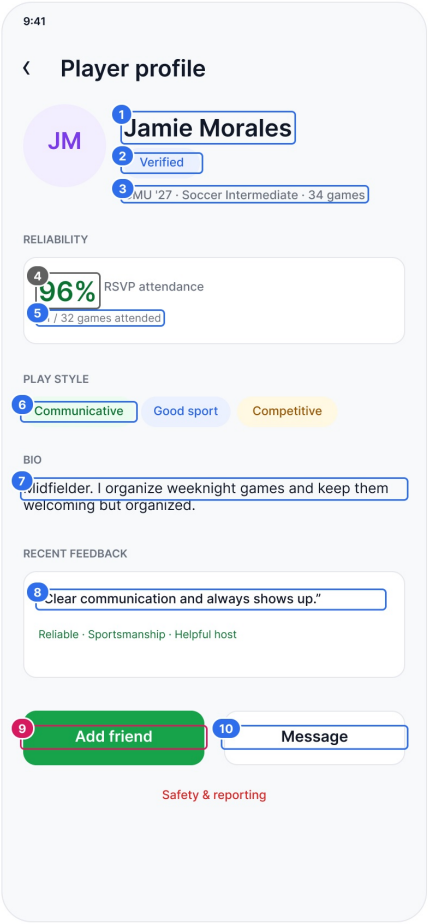
Endpoints: R1 · R8 · W16
Other: — | **Release: V1**



H09 — Game Detail

1. Storage · games.mediaURLs (Storage games/{id}/media)
2. Firestore · games/{id}.title
3. Firestore · games.visibility
4. Firestore · games.priceCents (Game.isFree)
5. Firestore · games.joinType
6. Firestore · games.skillLevel
7. Firestore · games.startTime / endTime
8. MapKit · games.venueName + distance to games.location
9. Open-Meteo · hour closest to startTime (HourlyForecast) + games.isIndoor
10. Firestore · games.surface, hasLights, fieldReserved, reservationWindow
11. Firestore · games.facilityNote
12. Firestore · games.joinedCount, playerLimit, spotsLeft
13. Firestore · games.attendeePreview
14. Computed · games/{id}/rsvps (approved) ∩ users/{me}/friends
15. Firestore · users/{hostId}.stats (PlayerStats.reliability, feedbackTagCounts)
16. Firestore · users/{hostId}.stats.avgResponseMinutes

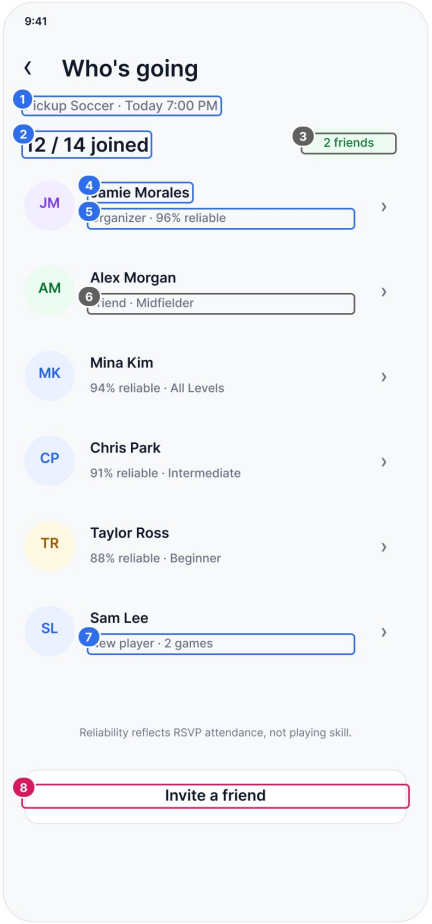
Endpoints: R2 · R3 · R4 · R7 · R9
Other: Storage · Open-Meteo · MapKit | **Release: V1**



H10 — Organizer Profile

1. Firestore · users/{id}.displayName
2. Firestore · users/{id}.isVerified
3. Firestore · schoolLine, sportLevels, gamesAttended + gamesHosted
4. Computed · PlayerStats.reliability = gamesAttended / gamesRSVPd
5. Firestore · stats.gamesAttended / gamesRSVPd
6. Firestore · top stats.feedbackTagCounts
7. Firestore · users/{id}.bio
8. Firestore · stats.recentFeedback
9. Input · users/{me}/friends/{id} + users/{id}/friends/{me} (Friendship)
10. Firestore · conversations (type direct, participantIds) – create if none

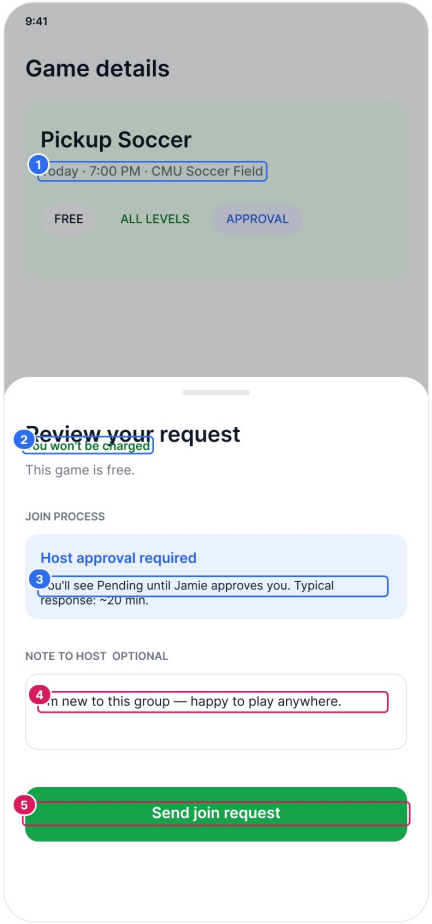
Endpoints: R7 · W12
Other: — | **Release: V1**



H11 — Player Roster

1. Firestore · games.title, startTime
2. Firestore · games.joinedCount / playerLimit
3. Computed · approved rsvps ∩ friends
4. Firestore · games/{id}/rsvps (approved): RSVP.displayName
5. Firestore · RSVP.isHost, RSVP.reliability
6. Computed · Friendship + users/{id}.position
7. Firestore · users/{id}.stats.gamesAttended
8. Input · users/{friendId}/notifications (newGame)

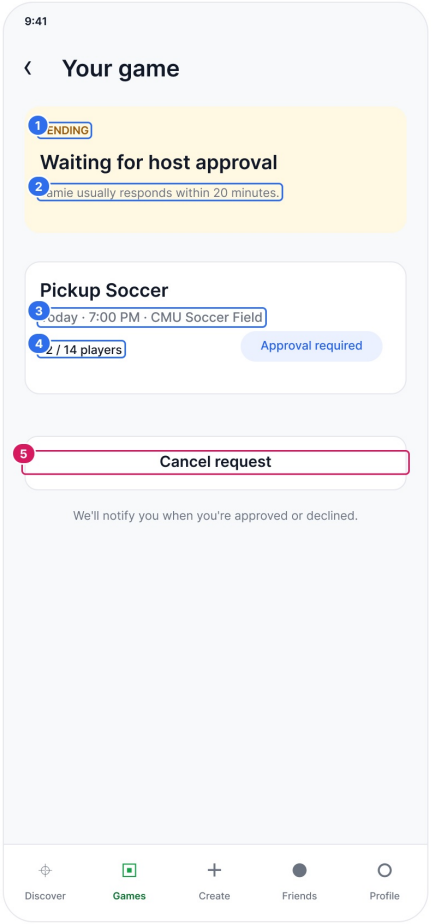
Endpoints: R2 · R3 · R7 · R9
Other: — | **Release: V1**



H12 — Join Review

1. Firestore · games.startTime, venueName
2. Firestore · games.priceCents
3. Firestore · games.joinType, hostName
4. Input · games/{id}/rsvps/{me}.noteToHost
5. Input · rsvps/{me} status pending, requestedAt; games.pendingCount += 1

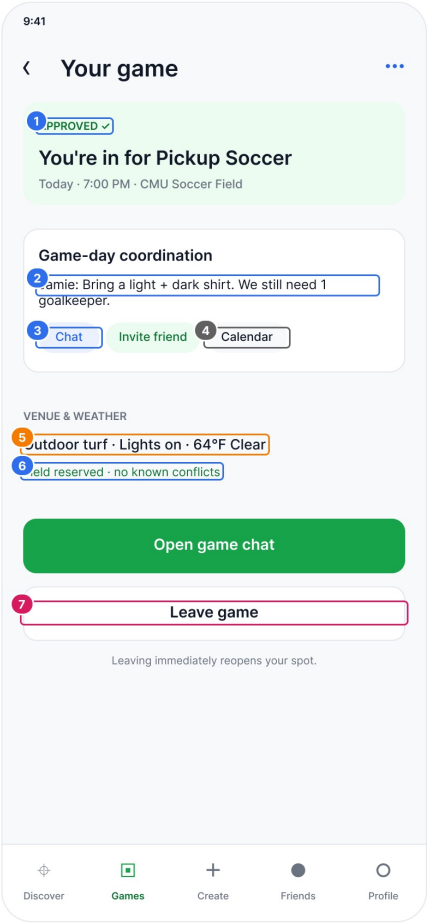
Endpoints: R2 · R4 · W6
Other: — | **Release: V1**



H13 — Pending Approval

1. Firestore · games/{id}/rsvps/{me}.status (live listener)
2. Firestore · users/{hostId}.stats.avgResponseMinutes
3. Firestore · games.startTime, venueName
4. Firestore · games.joinedCount / playerLimit
5. Input · rsvps/{me}.status = left; pendingCount -= 1

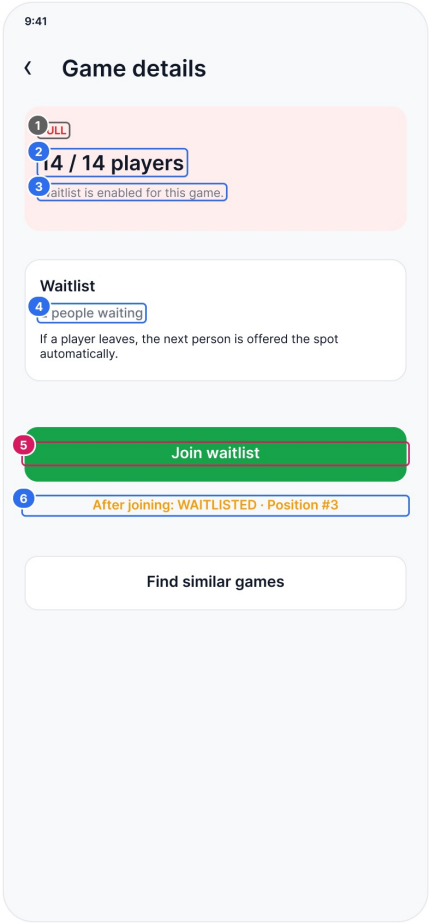
Endpoints: R2 · R4 · W8
Other: — | **Release: V1**



H14 — Joined Game

1. Firestore · rsvps/{me}.status
2. Firestore · games.notes
3. Firestore · games.conversationId → conversations
4. Computed · EventKit event from games.title, startTime, endTime
5. Open-Meteo · HourlyForecast + games.isIndoor, surface, hasLights
6. Firestore · games.fieldReserved, facilityNote
7. Input · opens H37 → rsvps/{me}.status = left

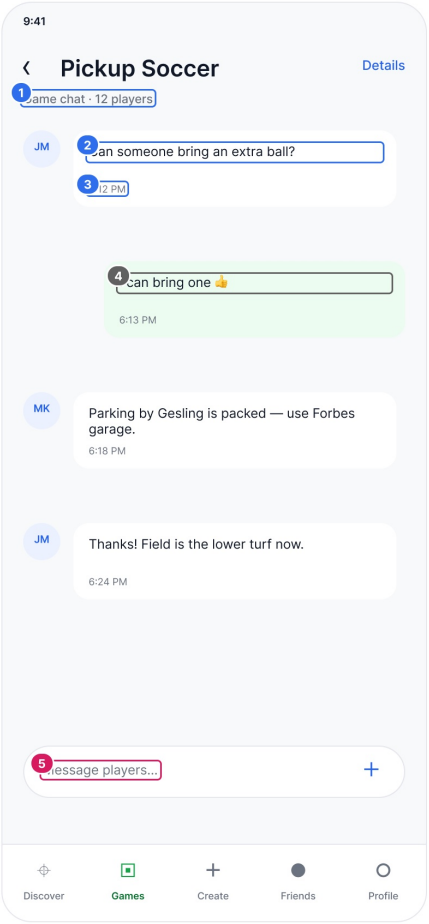
Endpoints: R2 · R4
Other: Open-Meteo · EventKit | **Release: V1**



H15 — Full / Waitlist

1. Computed · Game.isFull (joinedCount >= playerLimit)
2. Firestore · games.joinedCount / playerLimit
3. Firestore · games.waitlistEnabled
4. Firestore · games.waitlistCount
5. Input · rsvps/{me} status waitlisted; waitlistCount += 1
6. Firestore · RSVP.waitlistPosition

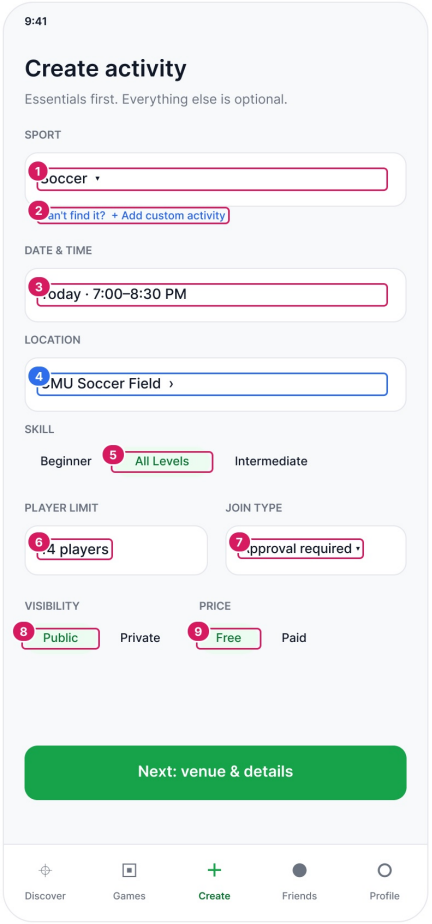
Endpoints: R1 · R2 · R4 · W6
Other: — | **Release: V1**



H16 — Event Chat

1. Firestore · conversations/{id}.title, participantIds.count
2. Firestore · conversations/{id}/messages (Message.text, sentAt) live
3. Firestore · Message.sentAt
4. Computed · own bubble when Message.senderId == me
5. Input · new messages doc + lastMessageText/At, unreadCounts

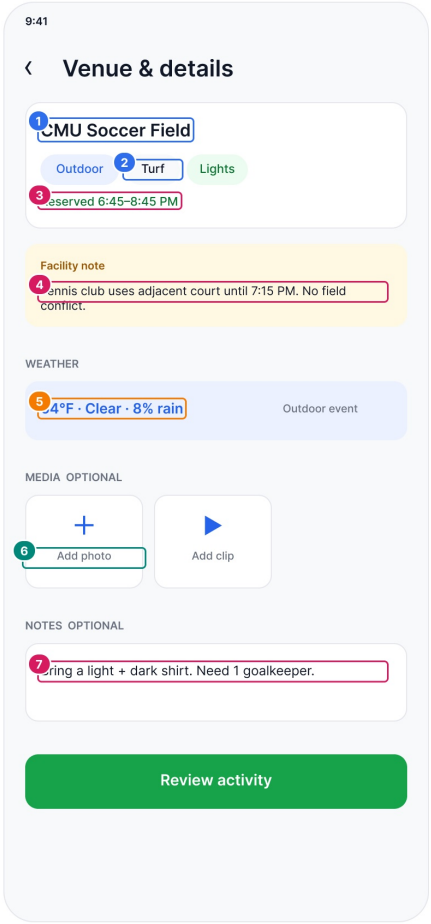
Endpoints: R13 · W11 · W17
Other: Storage | **Release: V1**



H17 — Create Activity

1. Input · games.sport
2. Input · games.customSportName
3. Input · games.startTime / endTime
4. Firestore · opens H40 → venueId, venueName, location, geohash
5. Input · games.skillLevel
6. Input · games.playerLimit
7. Input · games.joinType
8. Input · games.visibility
9. Input · games.priceCents

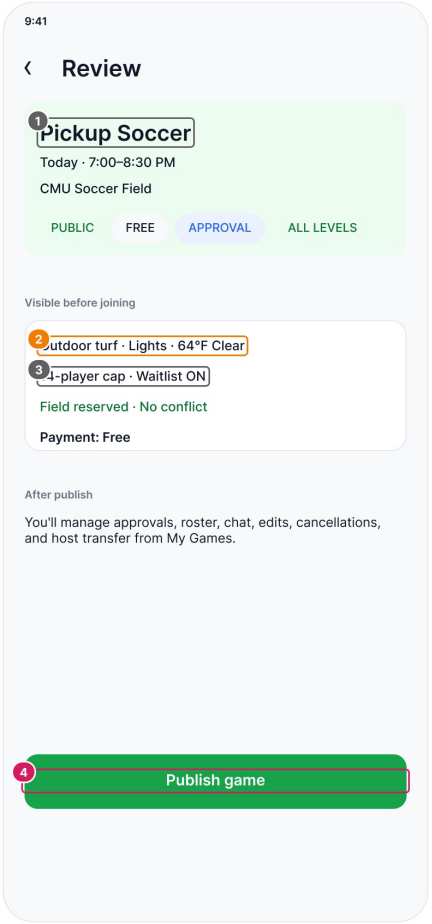
Endpoints: —
Other: — | Release: V1



H18 — Venue & Details

1. Firestore · venues/{id}.name
2. Firestore · venues.isIndoor, surface, hasLights → copied to games
3. Input · games.fieldReserved, reservationWindow
4. Input · games.facilityNote
5. Open-Meteo · temperature_2m, weather_code, precipitation_probability
6. Storage · Storage games/{id}/media → games.mediaURLs
7. Input · games.notes

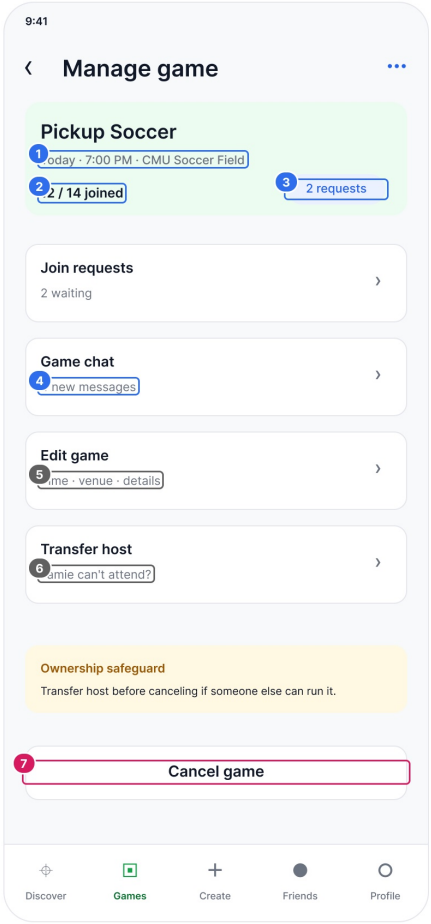
Endpoints: R11
Other: Storage · Open-Meteo | Release: V1



H19 — Review & Publish

1. Computed · draft Game from H17–H18 (not saved yet)
2. Open-Meteo · HourlyForecast + venue fields
3. Computed · draft playerLimit, waitlistEnabled
4. Input · new games doc + rsvps/{me} (isHost) + conversations (type game)

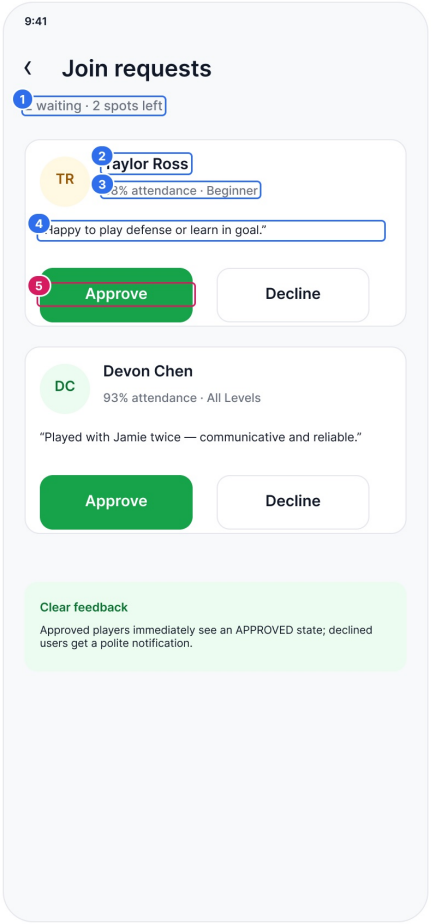
Endpoints: R2 · W3
Other: Open-Meteo | **Release: V1**



H20 — Host Dashboard

1. Firestore · games.startTime, venueName
2. Firestore · games.joinedCount / playerLimit
3. Firestore · games.pendingCount
4. Firestore · conversations/{id}.unreadCounts[me]
5. Computed · opens H46 Edit Game
6. Computed · opens H22 Transfer Host
7. Input · opens H38 → games.status = cancelled

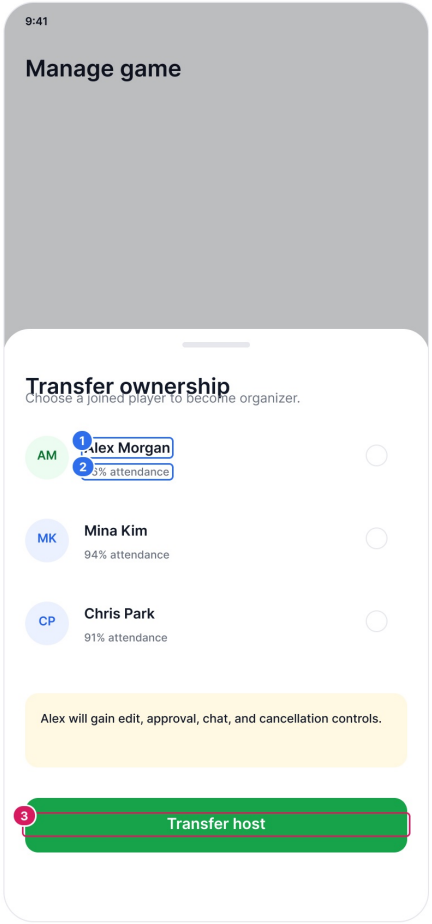
Endpoints: R2 · R6 · R12
Other: — | **Release: V1**



H21 — Join Requests

1. Firestore · games.pendingCount, Game.spotsLeft
2. Firestore · games/{id}/rsvps (pending): RSVP.displayName
3. Firestore · RSVP.reliability, RSVP.skillLevel
4. Firestore · RSVP.noteToHost
5. Input · RSVP.status approved/declined, respondedAt, hostResponse; counts; notification + FCM

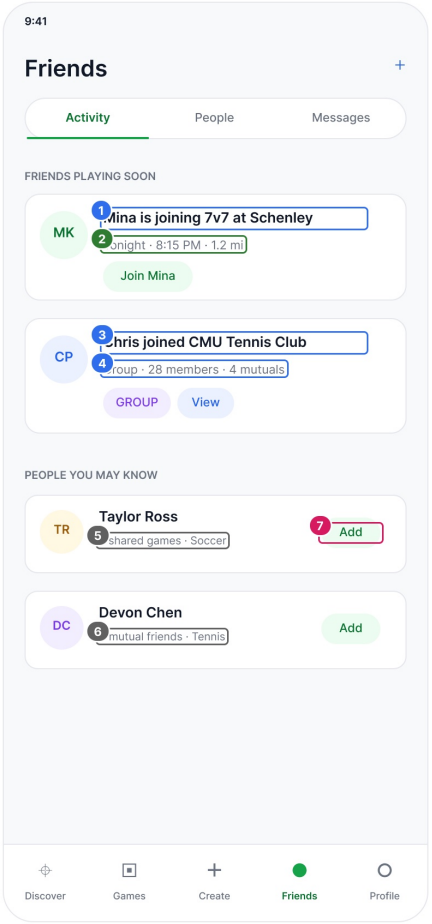
Endpoints: R2 · R3 · W7
Other: — | **Release: V1**



H22 — Transfer Host

1. Firestore · rsvps (approved, isHost false): RSVP.displayName
2. Firestore · RSVP.reliability
3. Input · games.hostId, hostName; RSVP.isHost swapped

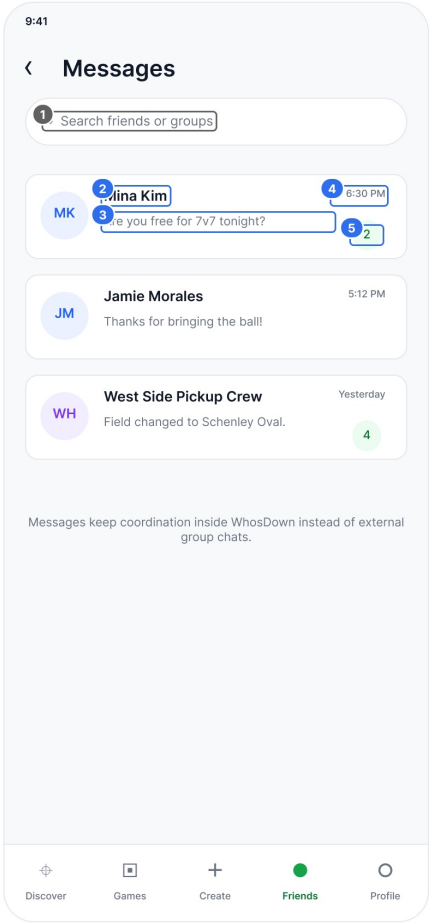
Endpoints: R2 · R3 · W10
Other: — | **Release: V2**



H23 — Friends Activity

1. **Firestore** · collectionGroup rsvps (userId in friendIds, approved) → games
2. **MapKit** · games.startTime + distance
3. **Firestore** · collectionGroup members (friend, recent joinedAt) → groups
4. **Firestore** · groups.memberCount + members ∩ friends
5. **Computed** · shared past rsvps
6. **Computed** · friends-of-friends (Friendship.mutualCount)
7. **Input** · users/{me}/friends/{id} requestSent

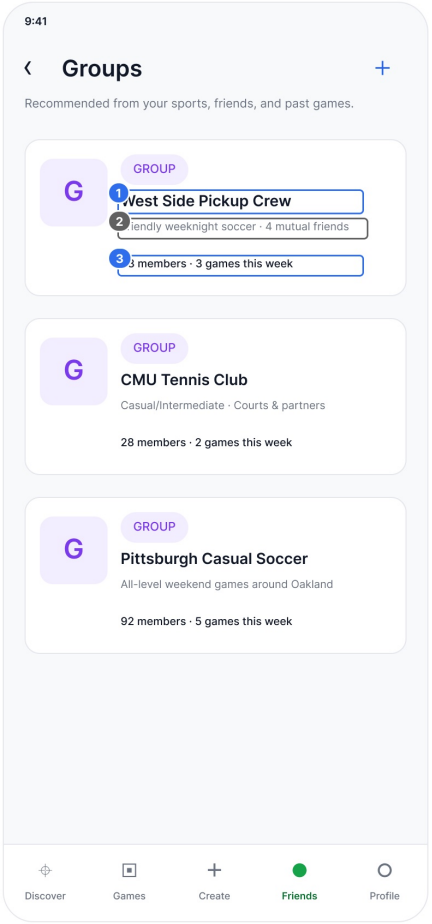
Endpoints: R9 · R17 · W12
Other: MapKit | **Release: V2**



H24 — Direct Messages

1. **Computed** · filter loaded friends + conversations
2. **Firestore** · conversations (participantIds contains me, by lastMessageAt): title
3. **Firestore** · Conversation.lastMessageText
4. **Firestore** · Conversation.lastMessageAt
5. **Firestore** · Conversation.unreadCounts[me]

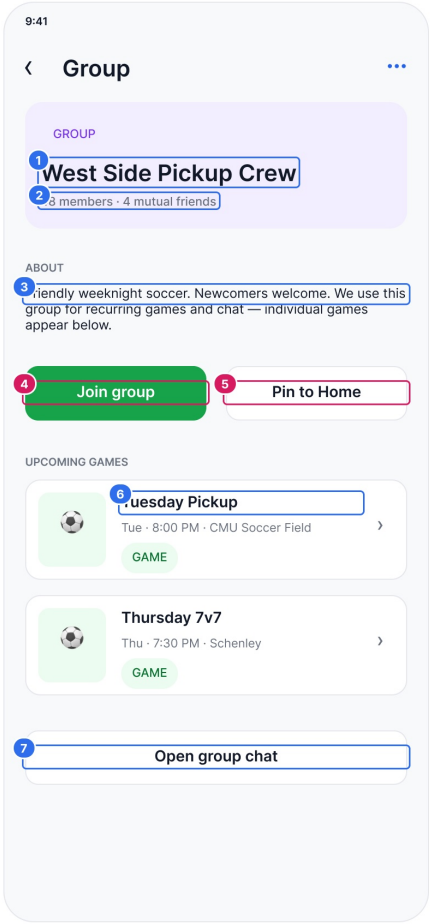
Endpoints: R9 · R12 · W17
Other: — | **Release: V2**



H25 — Groups Discover

1. **Firestore** · groups (public, sports array-contains-any): name
2. **Computed** · groups.description + members ∩ friends
3. **Firestore** · groups.memberCount + count(games where groupId, this week)

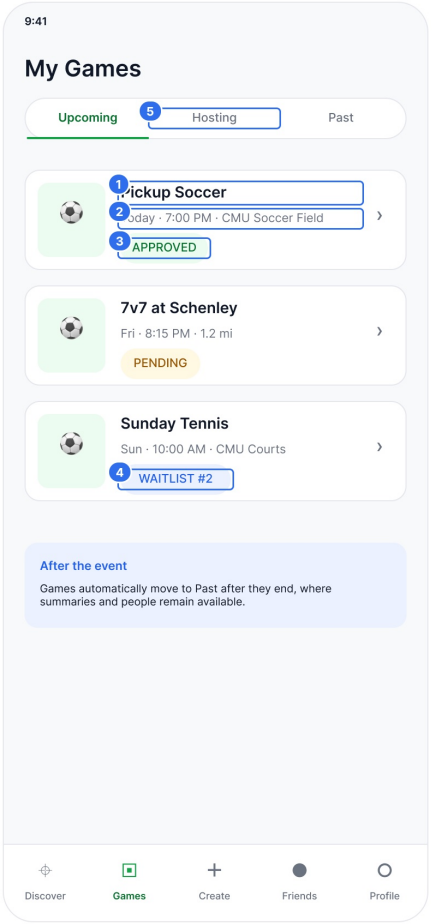
Endpoints: R9 · R14 · R15 · R16
Other: — | **Release: V2**



H26 — Group Detail

1. **Firestore** · groups/{id}.name
2. **Firestore** · groups.memberCount + members ∩ friends
3. **Firestore** · groups.description
4. **Input** · groups/{id}/members/{me} (member or pending); memberCount += 1
5. **Input** · private/settings.pinnedGroupIds
6. **Firestore** · games where groupId == id, startTime >= now
7. **Firestore** · groups.conversationId → conversations

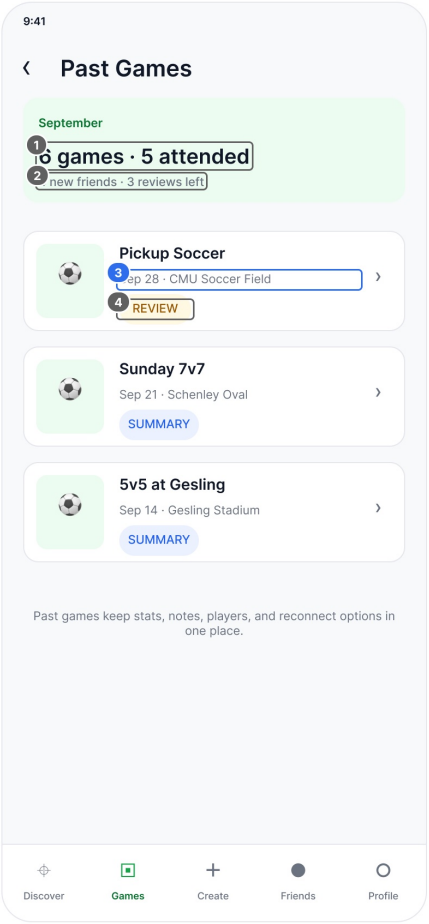
Endpoints: R14 · R15 · R16 · W13 · W16
Other: — | **Release: V2**



H27 — My Games

1. Firestore · collectionGroup rsvps (userId == me) → games.title
2. Firestore · games.startTime, venueName
3. Firestore · RSVP.status
4. Firestore · RSVP.waitlistPosition
5. Firestore · games where hostId == me

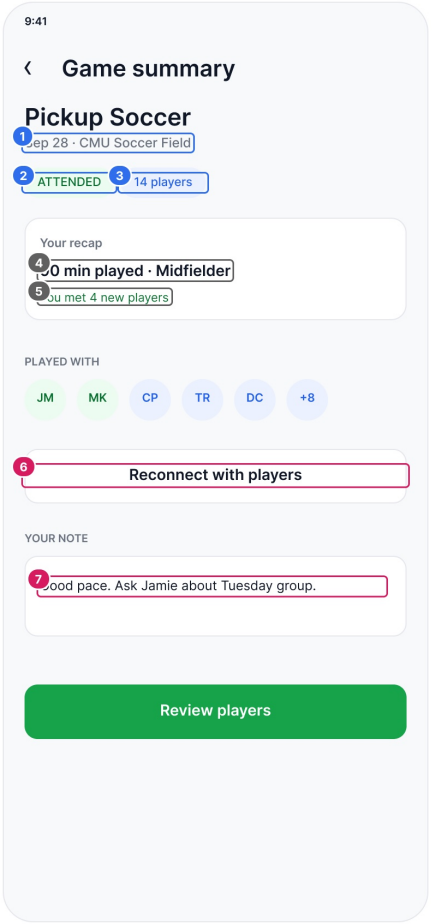
Endpoints: R5 · R6
Other: — | **Release: V1**



H28 — Past Games

1. Computed · past rsvps + RSVP.attended
2. Computed · Friendship.since this month + reviews by me
3. Firestore · games.startTime, venueName
4. Computed · no reviews doc yet for this game

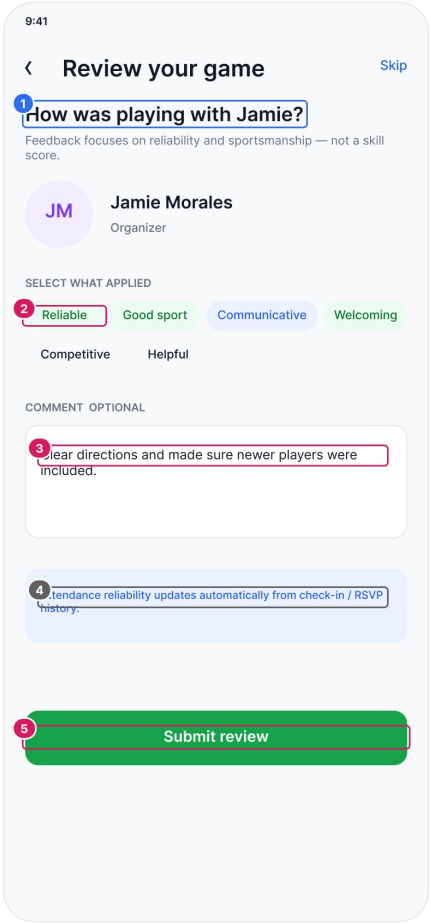
Endpoints: R5 · R18
Other: — | **Release: V1**



H29 — Past Game Summary

1. Firestore · games.startTime, venueName
2. Firestore · rsvps/{me}.attended
3. Firestore · games.joinedCount
4. Computed · endTime − startTime + users/{me}.position
5. Computed · approved rsvps minus friends
6. Input · Friendship request
7. Input · rsvps/{me}.personalNote

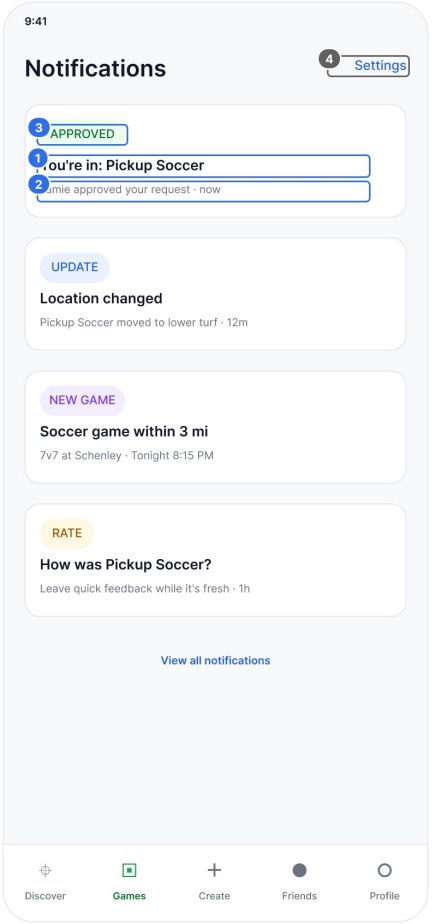
Endpoints: R2 · R3 · R4 · R7 · R9 · W12 · W19
Other: — | **Release: V2**



H30 — Post-game Review

1. Firestore · rsvps: displayName, isHost
2. Input · reviews.tags ([FeedbackTag])
3. Input · reviews.comment
4. Computed · PlayerStats.reliability
5. Input · new reviews doc

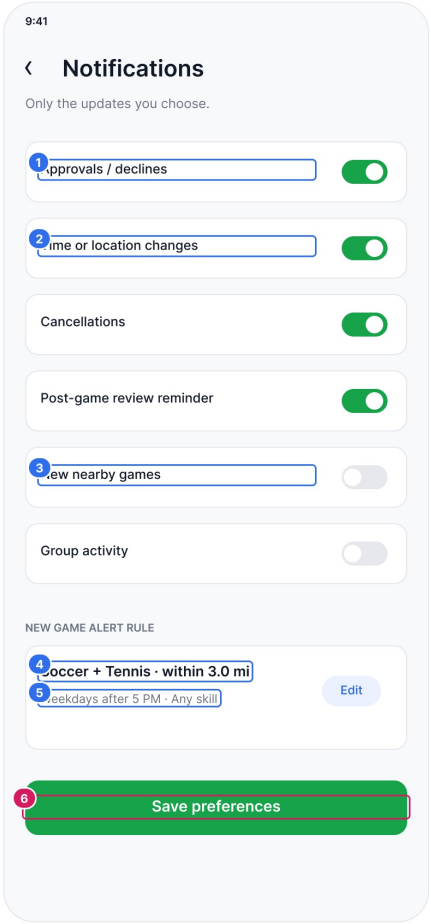
Endpoints: R3 · W14
Other: — | **Release: V2**



H31 — Notifications

1. **Firestore** · users/{me}/notifications (by createdAt): title
2. **Firestore** · AppNotification.body, createdAt
3. **Firestore** · AppNotification.type
4. **Computed** · opens H32

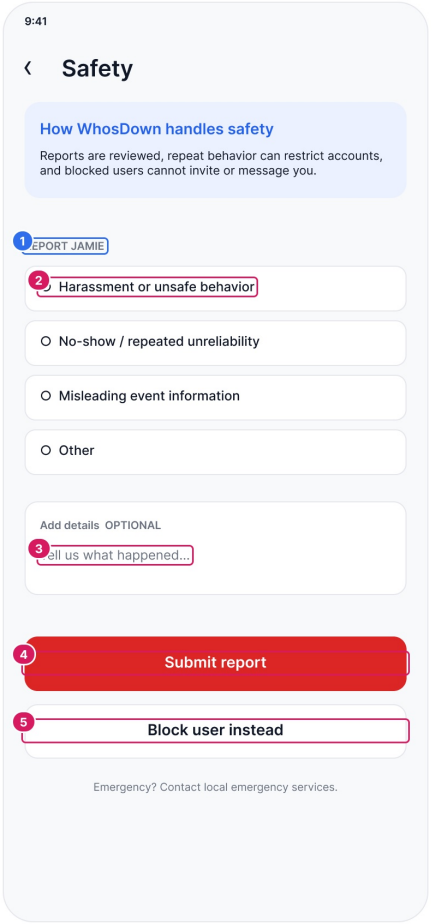
Endpoints: R10 · W17
Other: — | **Release: V1**



H32 — Notification Settings

1. **Firestore** · notificationPrefs.approvals
2. **Firestore** · notificationPrefs.timeOrLocationChanges
3. **Firestore** · notificationPrefs.newNearbyGames
4. **Firestore** · newGameAlert.sports, radiusMiles
5. **Firestore** · newGameAlert.weekdaysOnly, earliestHour, skillLevels
6. **Input** · write private/settings

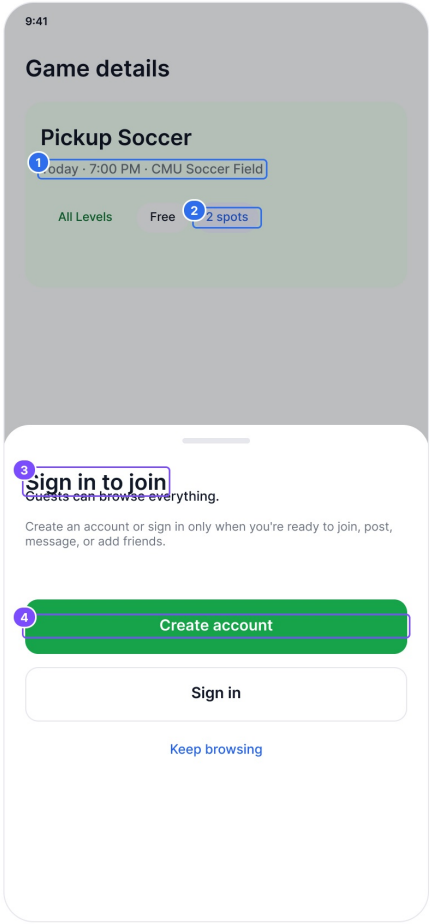
Endpoints: R8 · W16
Other: — | **Release: V2**



H33 — Safety & Reporting

1. **Firestore** · users/{id}.displayName
2. **Input** · reports.reason (ReportReason)
3. **Input** · reports.details
4. **Input** · new reports doc (status open)
5. **Input** · private/settings.blockedUserIds

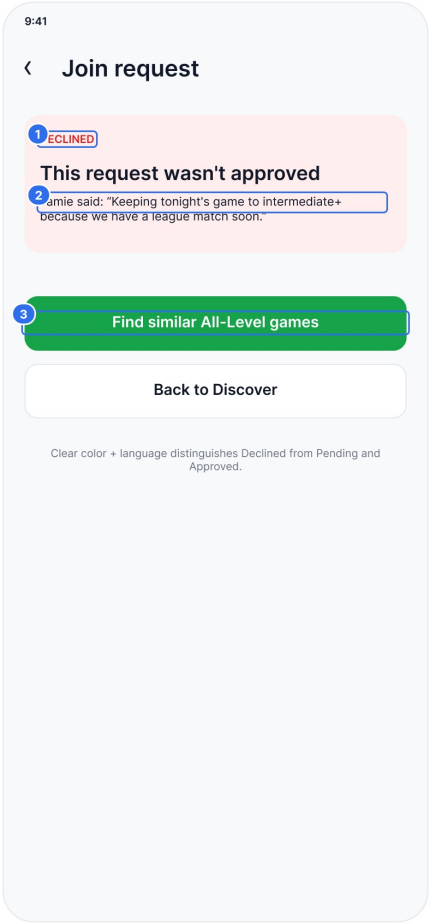
Endpoints: R8 · W15
Other: — | **Release: V1 report · V2 block**



H34 — Guest Join Gate

1. **Firestore** · games.startTime, venueName
2. **Firestore** · Game.spotsLeft
3. **Auth** · shown when Auth user is anonymous (AppUser.isGuest)
4. **Auth** · link anonymous account to email credential

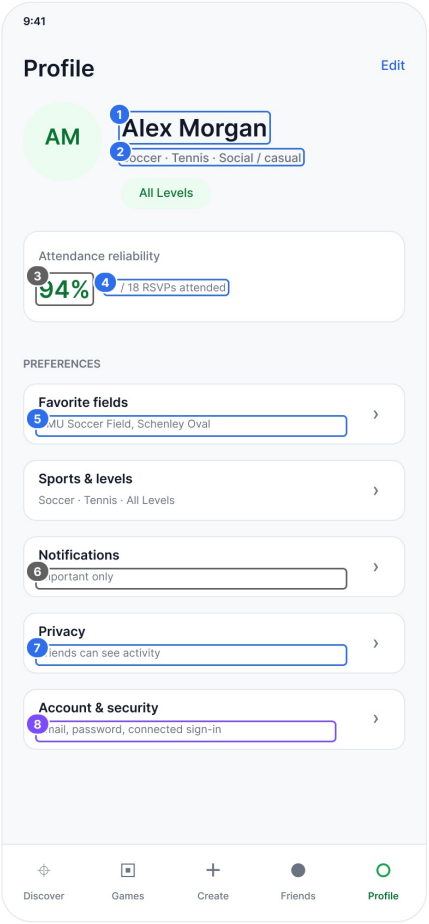
Endpoints: R2
Other: Auth | **Release: V1**



H35 — Declined Request

1. **Firestore** · rsvps/{me}.status
2. **Firestore** · RSVP.hostResponse
3. **Firestore** · nearby games where skillLevel == allLevels

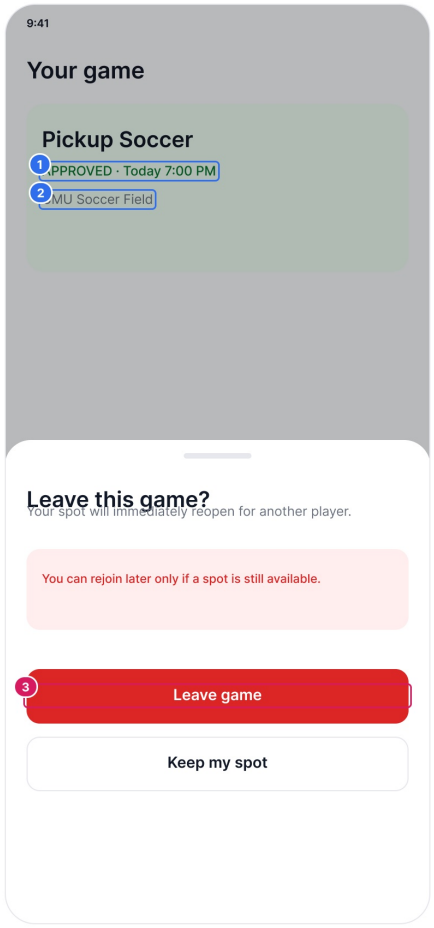
Endpoints: R1 · R4
Other: — | **Release: V1**



H36 — Profile & Preferences

1. **Firestore** · users/{me}.displayName
2. **Firestore** · users.sportLevels, playStyles
3. **Computed** · PlayerStats.reliability
4. **Firestore** · stats.gamesAttended / gamesRSVPd
5. **Firestore** · private/settings.favoriteVenueIds → venues.name
6. **Computed** · summary of notificationPrefs
7. **Firestore** · privacy.gameActivity
8. **Auth** · opens H59

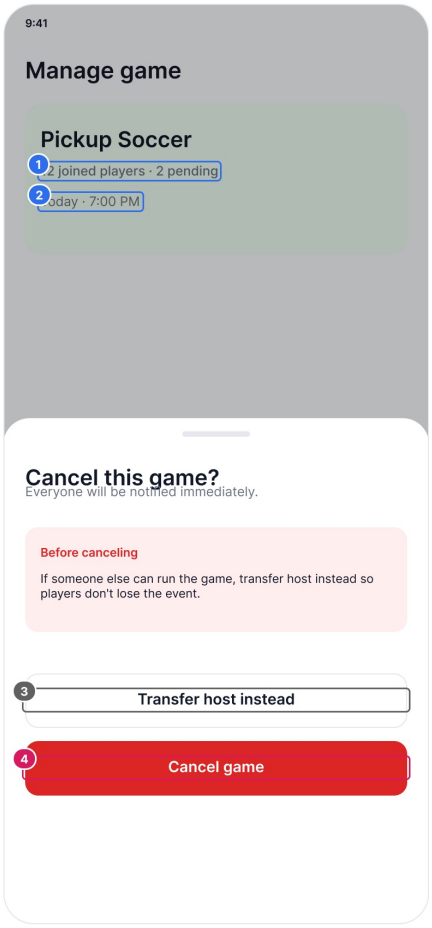
Endpoints: R7 · R8
Other: — | **Release: V1**



H37 — Leave Confirmation

1. **Firestore** · rsvps/{me}.status + games.startTime
2. **Firestore** · games.venueName
3. **Input** · rsvps/{me}.status = left; joinedCount -= 1; first waitlisted gets spotOfferExpiresAt + spotOpen notification

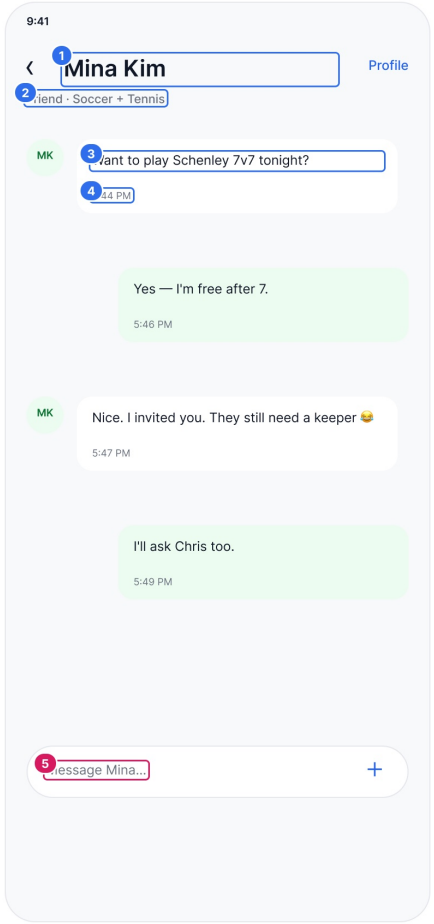
Endpoints: R2 · R4 · W8
Other: — | **Release: V1**



H38 — Cancel Game Confirmation

1. **Firestore** · games.joinedCount, pendingCount
2. **Firestore** · games.startTime
3. **Computed** · opens H22
4. **Input** · games.status = cancelled, cancelReason; gameCancelled notification to every RSVP + FCM

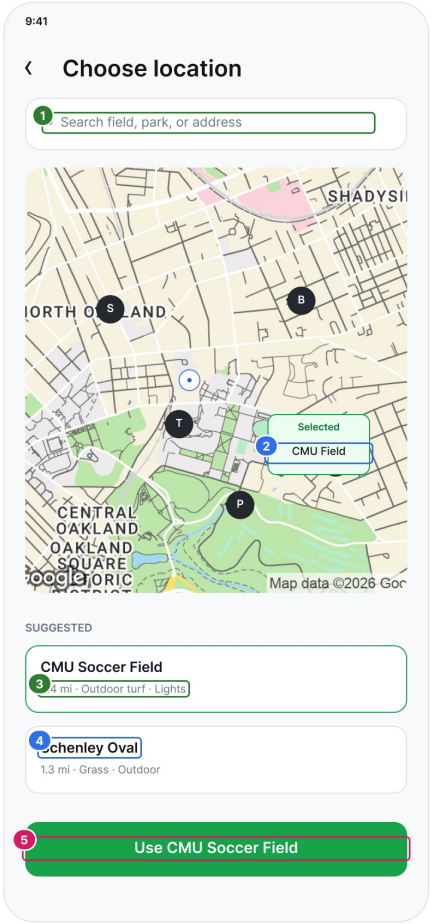
Endpoints: R2 · W5
Other: — | **Release: V1**



H39 — Message Thread

1. Firestore · users/{otherId}.displayName
2. Firestore · Friendship.status + users.sportLevels
3. Firestore · conversations/{id}/messages (type direct)
4. Firestore · Message.sentAt
5. Input · new messages doc + lastMessageText/At

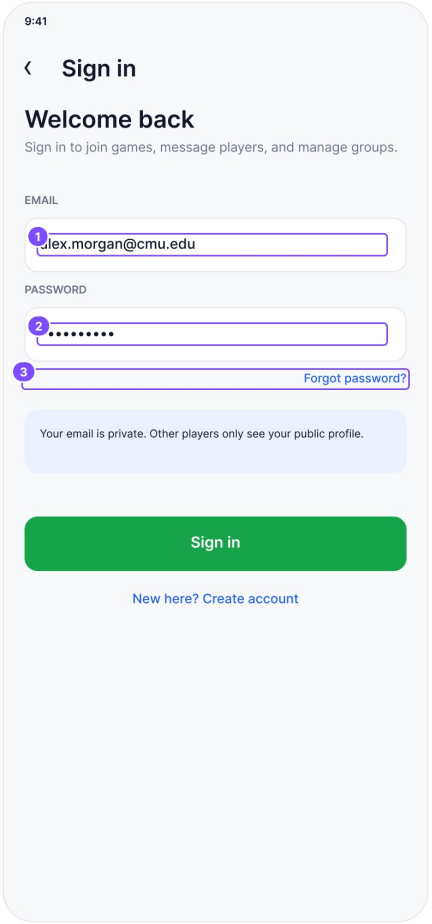
Endpoints: R7 · R13 · W11
Other: — | **Release: V2**



H40 — Choose Location

1. MapKit · MKLocalSearch + venues by geohash/name
2. Firestore · venues/{id}.name
3. MapKit · distance + venues.isIndoor, surface, hasLights
4. Firestore · favorites first (favoriteVenueIds), then nearby venues
5. Input · sets games.venueId, venueName, location, geohash (new venues doc if from MapKit)

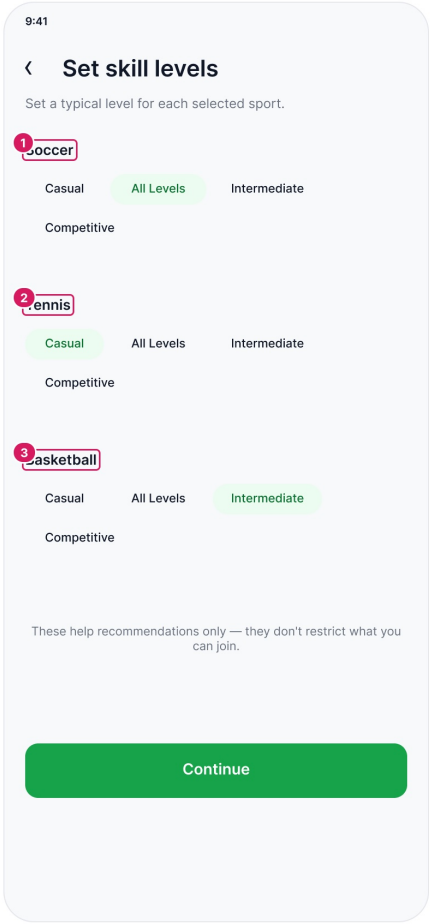
Endpoints: R11 · W18
Other: MapKit | **Release: V1**



H41 — Sign In

1. Auth · signIn(withEmail:password:)
2. Auth · password checked by Firebase Auth
3. Auth · sendPasswordReset(withEmail:)

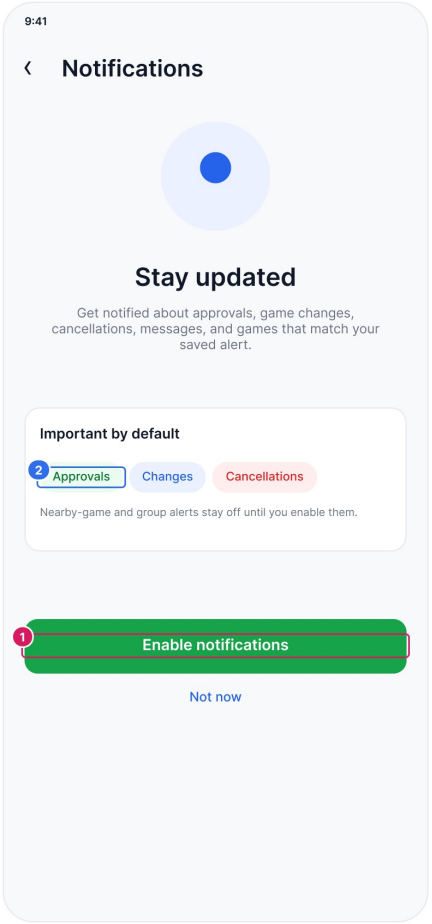
Endpoints: —
Other: Auth | **Release: V1**



H42 — Skill Levels

1. Input · users/{uid}.sportLevels["soccer"]
2. Input · users/{uid}.sportLevels["tennis"]
3. Input · users/{uid}.sportLevels["basketball"]

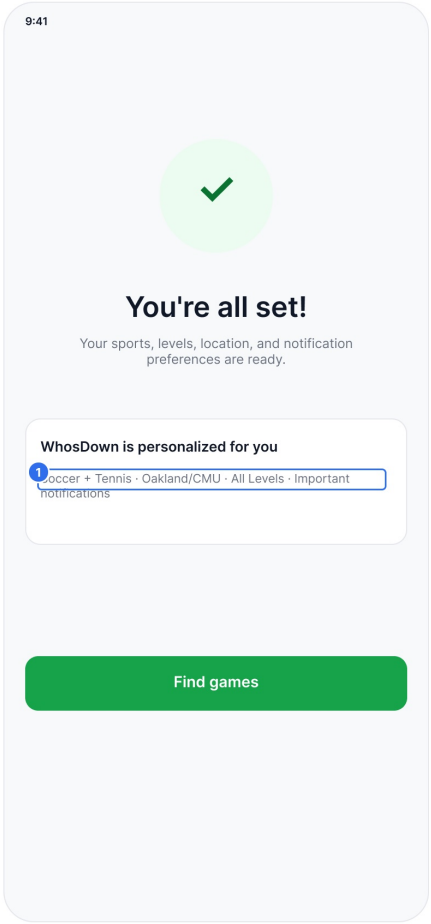
Endpoints: W1
Other: — | **Release: V1**



H43 — Notification Permission

1. Input · UNUserNotificationCenter + FCM token → private/settings.fcmTokens
2. Firestore · notificationPrefs defaults

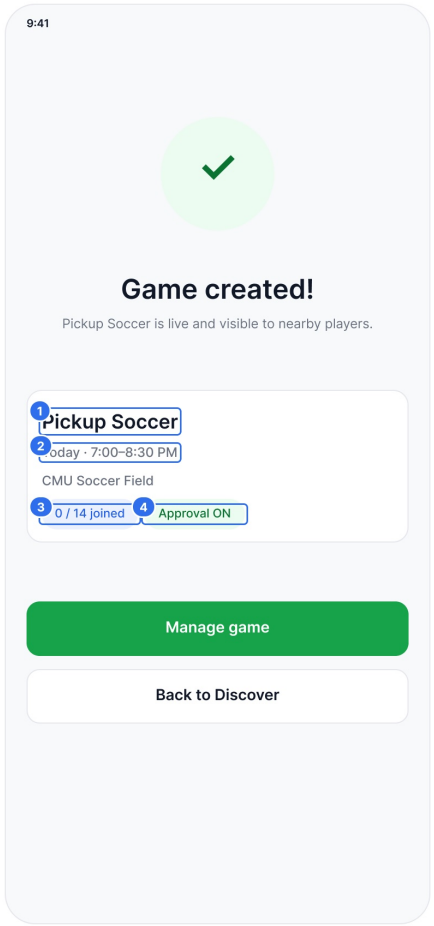
Endpoints: W1 · W16
Other: FCM | **Release: V2**



H44 — Onboarding Complete

1. Firestore · users.sportLevels, homeArea + notificationPrefs

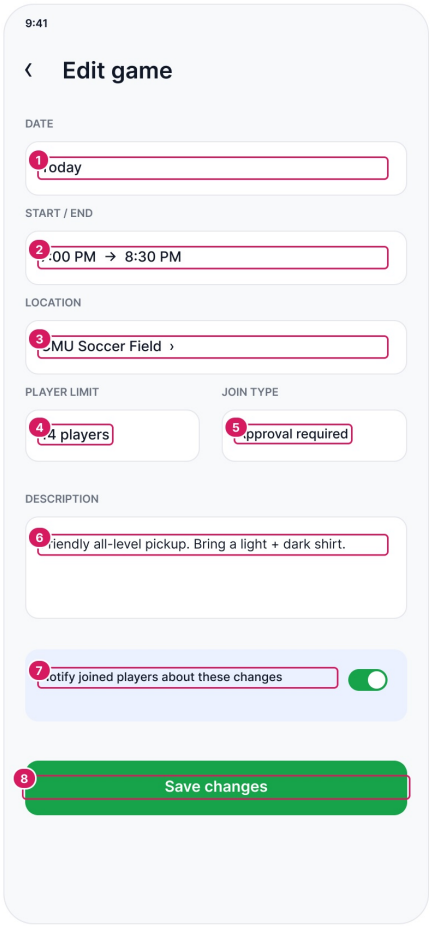
Endpoints: R7 · R8 · W1
Other: — | **Release: V1**



H45 — Game Created

1. Firestore · games/{id}.title (just written)
2. Firestore · games.startTime / endTime
3. Firestore · games.joinedCount / playerLimit
4. Firestore · games.joinType

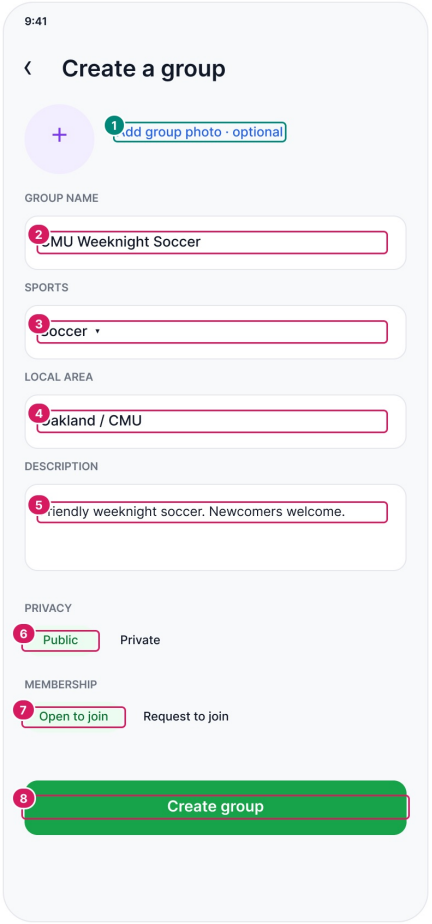
Endpoints: R2
Other: — | **Release: V1**



H46 — Edit Game

1. Input · games.startTime (date)
2. Input · games.startTime / endTime
3. Input · venueId, venueName, location (via H40)
4. Input · games.playerLimit
5. Input · games.joinType
6. Input · games.description
7. Input · users/{playerId}/notifications gameUpdated with change (GameChange)
8. Input · update games doc + updatedAt

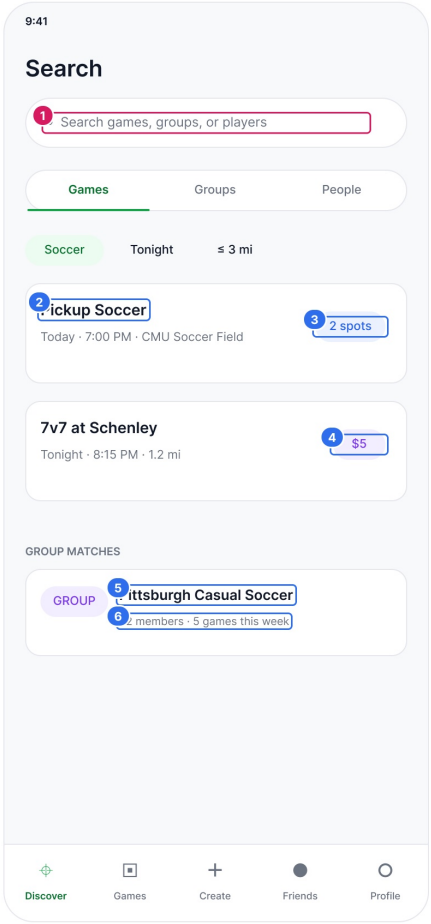
Endpoints: R2 · W4
Other: — | **Release: V1**



H47 — Create Group

1. **Storage** · Storage groups/{id}/photo.jpg → photoURL
2. **Input** · groups.name (+ nameLowercase)
3. **Input** · groups.sports
4. **Input** · groups.localArea
5. **Input** · groups.description
6. **Input** · groups.visibility
7. **Input** · groups.membershipType
8. **Input** · new groups doc + members/{me} owner + conversations (type group)

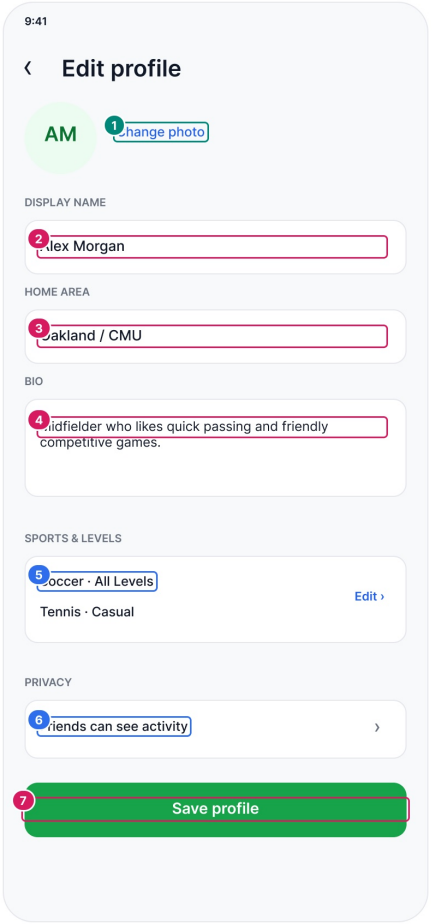
Endpoints: W13
Other: Storage | **Release: V2**



H48 — Search

1. **Input** · prefix query on titleLowercase / nameLowercase / displayNameLowercase
2. **Firestore** · games.title
3. **Firestore** · Game.spotsLeft
4. **Firestore** · games.priceCents
5. **Firestore** · groups.name
6. **Firestore** · groups.memberCount + count(games)

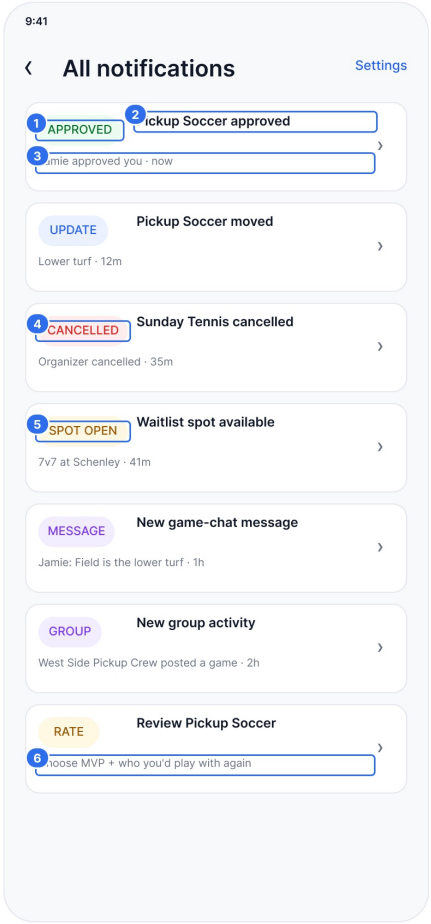
Endpoints: R16 · R19
Other: — | **Release: V2**



H49 — Edit Profile

1. **Storage** · Storage users/{uid}/avatar.jpg → photoURL
2. **Input** · users.displayName
3. **Input** · users.homeArea
4. **Input** · users.bio
5. **Firestore** · users.sportLevels (edit opens H42)
6. **Firestore** · privacy.gameActivity (opens H57)
7. **Input** · update users doc

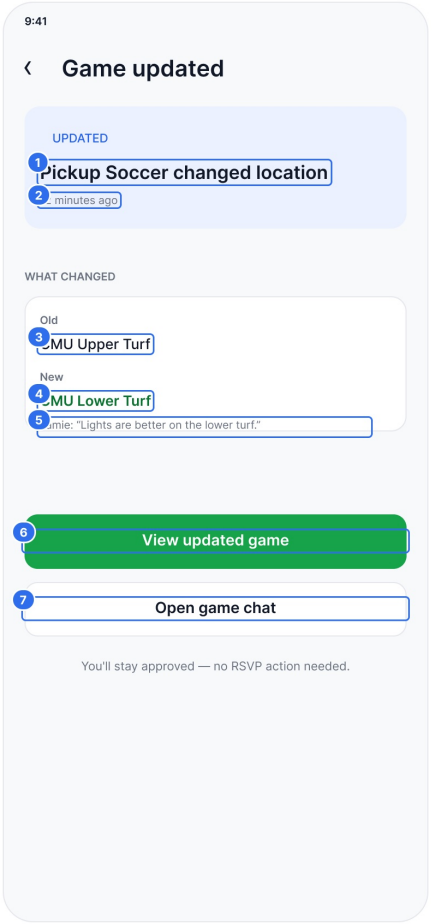
Endpoints: R7 · W2
Other: Storage | **Release: V1**



H50 — All Notifications

1. **Firestore** · users/{me}/notifications (by createdAt): AppNotification.type
2. **Firestore** · AppNotification.title
3. **Firestore** · AppNotification.body, createdAt
4. **Firestore** · type gameCancelled
5. **Firestore** · type spotOpen
6. **Firestore** · type reviewReminder

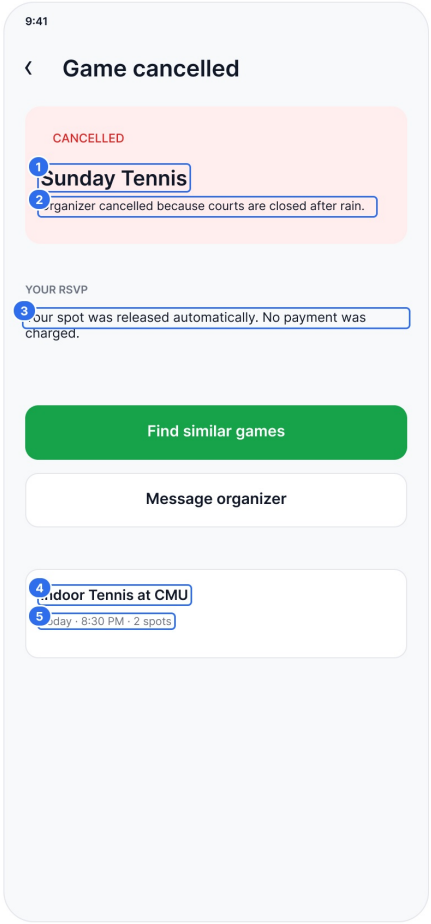
Endpoints: R10 · W17
Other: FCM | **Release: V1**



H51 — Game Update Notification

1. Firestore · AppNotification.title
2. Firestore · AppNotification.createdAt
3. Firestore · GameChange.oldValue
4. Firestore · GameChange.newValue
5. Firestore · GameChange.hostNote
6. Firestore · AppNotification.gameId → games
7. Firestore · games.conversationId

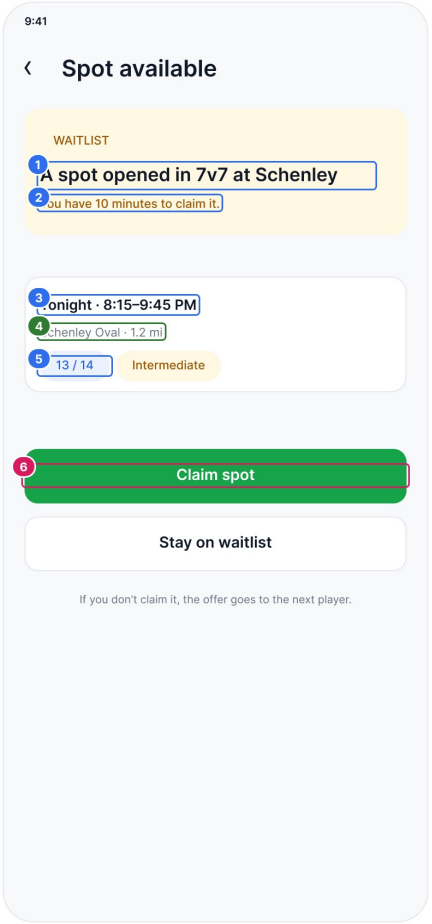
Endpoints: R2 · R10 · W17
Other: FCM | **Release: V1**



H52 — Game Cancelled Notification

1. Firestore · games.title, status cancelled
2. Firestore · games.cancelReason
3. Firestore · RSVP.status + games.priceCents
4. Firestore · nearby games, same sport
5. Firestore · games.startTime, Game.spotsLeft

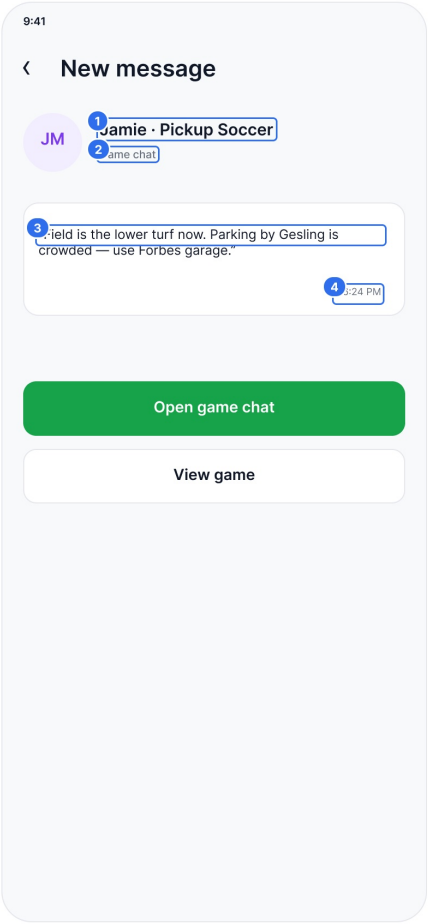
Endpoints: R1 · R2 · R10 · W17
Other: FCM | **Release: V1**



H53 — Waitlist Spot Available

1. **Firestore** · AppNotification (spotOpen) + games.title
2. **Firestore** · RSVP.spotOfferExpiresAt / AppNotification.expiresAt
3. **Firestore** · games.startTime / endTime
4. **MapKit** · games.venueName + distance
5. **Firestore** · games.joinedCount / playerLimit
6. **Input** · RSVP.status approved; joinedCount += 1; waitlistCount -= 1

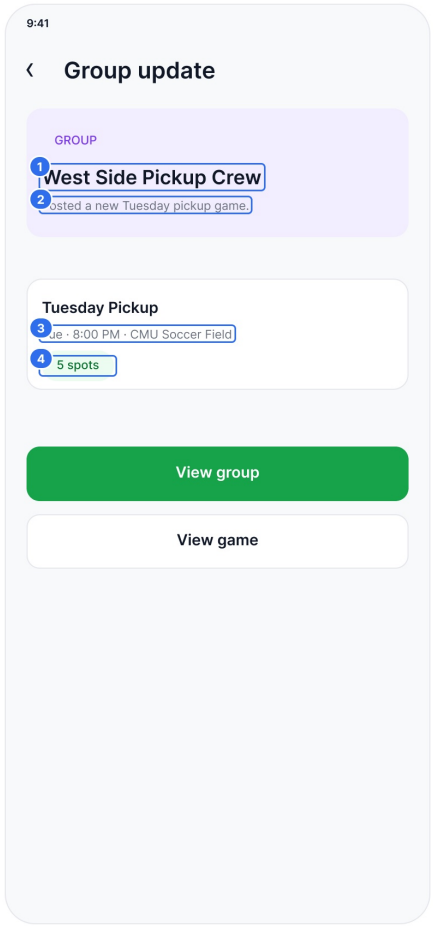
Endpoints: R2 · R4 · R10 · W9 · W17
Other: FCM · MapKit | **Release: V1**



H54 — Message Notification

1. **Firestore** · Message.senderName + Conversation.title
2. **Firestore** · Conversation.type
3. **Firestore** · Message.text
4. **Firestore** · Message.sentAt

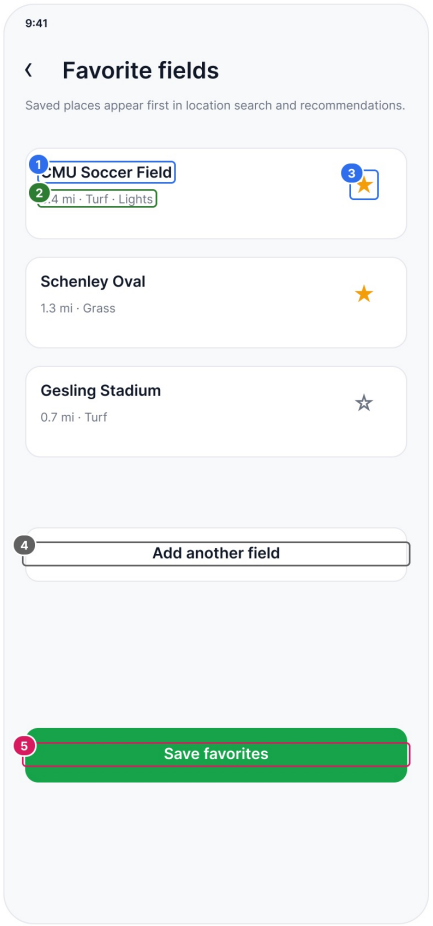
Endpoints: R10 · R13 · W17
Other: FCM | **Release: V2**



H55 — Group Activity Notification

1. Firestore · AppNotification.groupId → groups.name
2. Firestore · AppNotification.body
3. Firestore · games.startTime, venueName
4. Firestore · Game.spotsLeft

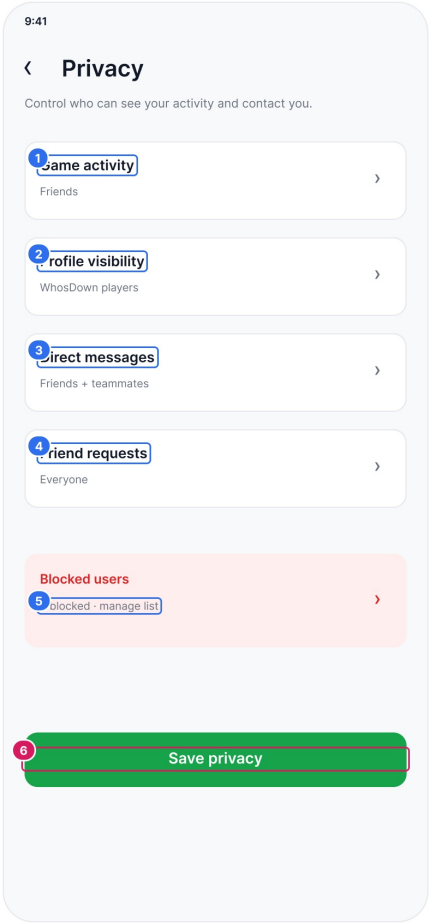
Endpoints: R2 · R10 · R14 · W17
Other: FCM | **Release: V2**



H56 — Favorite Fields

1. Firestore · venues.name
2. MapKit · distance + venues.surface, hasLights
3. Firestore · private/settings.favoriteVenueIds
4. Computed · opens H40
5. Input · write favoriteVenueIds

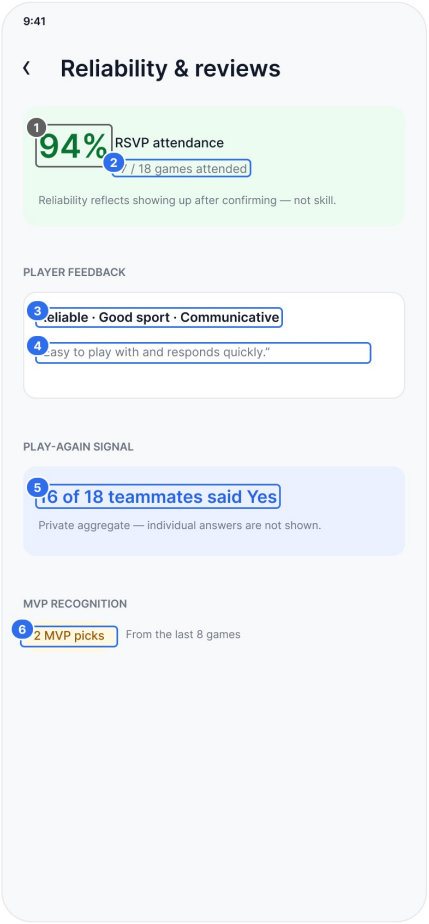
Endpoints: R8 · R11 · W16
Other: MapKit | **Release: V2**



H57 — Privacy Settings

1. Firestore · privacy.gameActivity (AudienceSetting)
2. Firestore · privacy.profileVisibility
3. Firestore · privacy.directMessages
4. Firestore · privacy.friendRequests
5. Firestore · blockedUserIds.count
6. Input · write private/settings.privacy

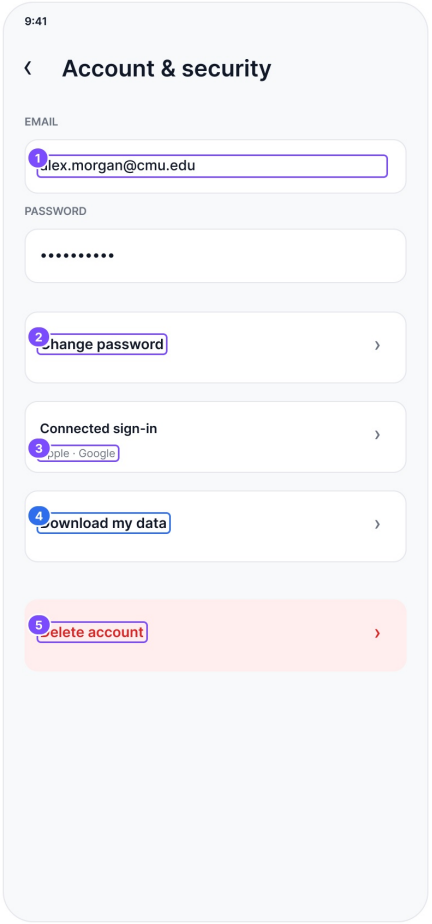
Endpoints: R8 · W15 · W16
Other: — | **Release: V2**



H58 — Reliability & Reviews

1. Computed · PlayerStats.reliability
2. Firestore · stats.gamesAttended / gamesRSVPd
3. Firestore · stats.feedbackTagCounts
4. Firestore · stats.recentFeedback
5. Firestore · stats.playAgainYes / playAgainTotal
6. Firestore · stats.mvpCount

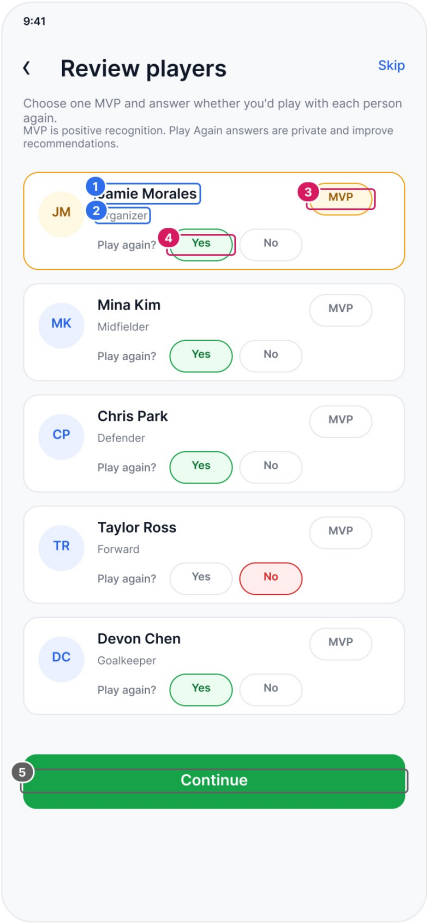
Endpoints: R7
Other: — | **Release: V2**



H59 — Account & Security

1. **Auth** · Auth currentUser.email (also private/ settings.email)
2. **Auth** · updatePassword(to:)
3. **Auth** · currentUser.providerData
4. **Firestore** · export of the user's documents (V2)
5. **Auth** · currentUser.delete() + remove user docs

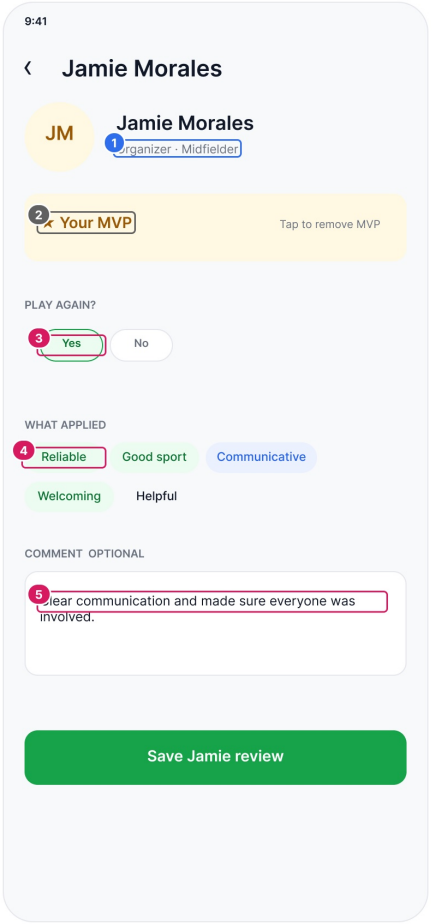
Endpoints: R8
Other: Auth | **Release: V2**



H60 — Review Players

1. **Firestore** · games/{id}/rsvps (attended): displayName
2. **Firestore** · RSVP.isHost / users.position
3. **Input** · reviews.isMVP (one per game)
4. **Input** · reviews.playAgain
5. **Computed** · draft reviews kept on device until H62

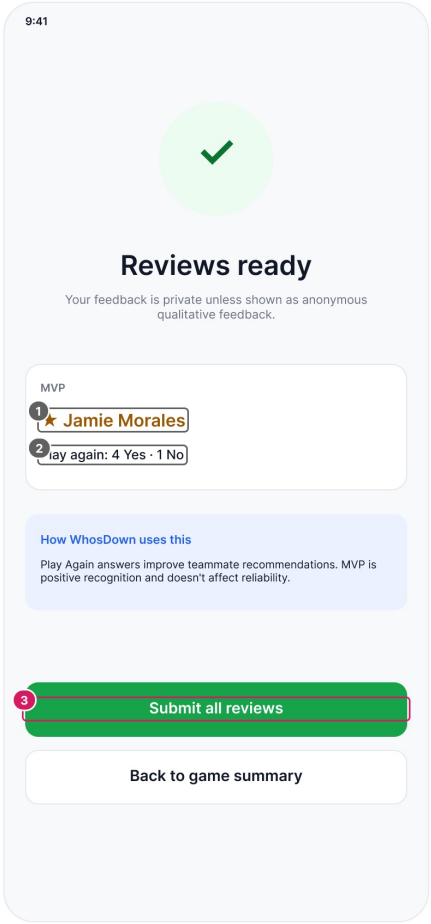
Endpoints: R3 · R7 · W14
Other: — | **Release: V2**



H61 — Player Review Detail

1. **Firestore** · RSVP.isHost + users.position
2. **Computed** · draft Review.isMVP
3. **Input** · Review.playAgain
4. **Input** · Review.tags
5. **Input** · Review.comment

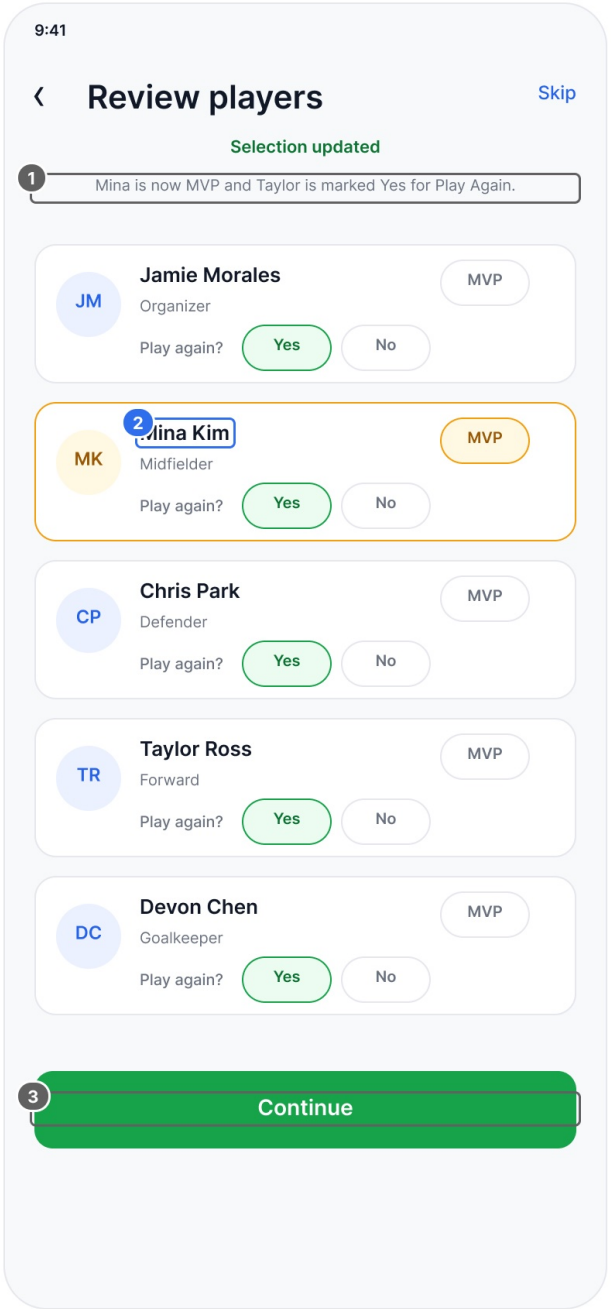
Endpoints: R3 · R7 · W14
Other: — | **Release: V2**



H62 — Review Summary

1. **Computed** · draft Review.isMVP
2. **Computed** · count of draft Review.playAgain
3. **Input** · one reviews doc per teammate; reviewee stats (mvpCount, playAgainYes/Total, feedbackTagCounts) incremented

Endpoints: W14
Other: — | **Release: V2**



H63 — Review Players Changed

1. **Computed** · updated draft Review.isMVP / playAgain
2. **Firestore** · rsvps: displayName
3. **Computed** · drafts kept until submit

Endpoints: R3
Other: — | **Release: V2**